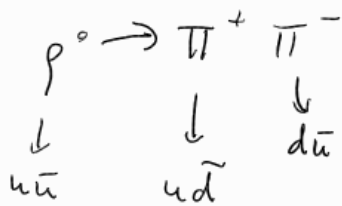
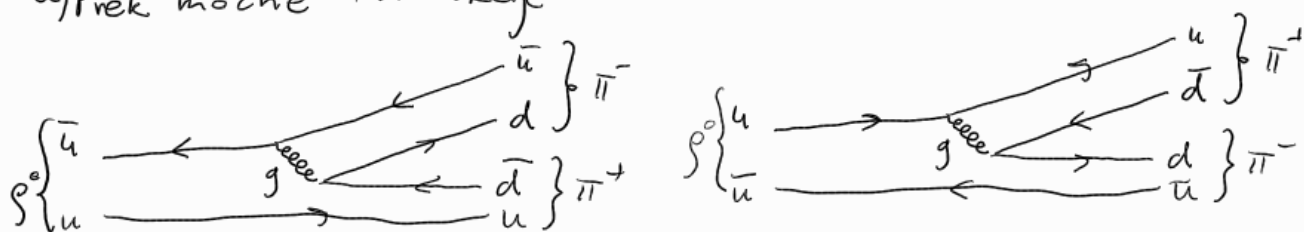


Nariši F. D. za razpad

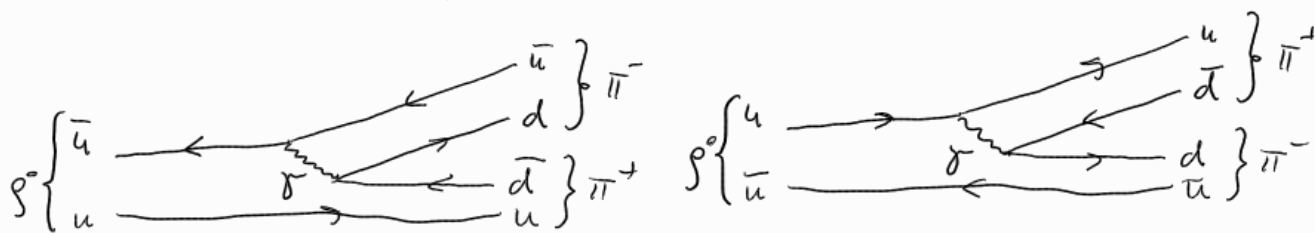


Pogoj za potek reakcije: $m_{\rho^0} c^2 > 2m_{\pi^\pm} c^2$

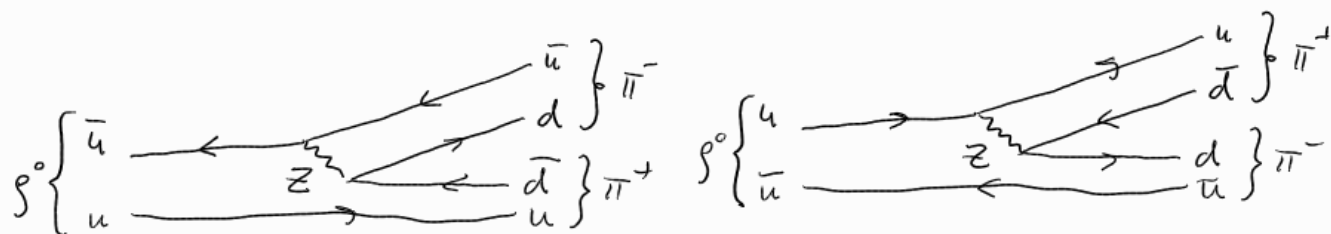
a) Prek močne interakcije



b) Prek EM interakcije

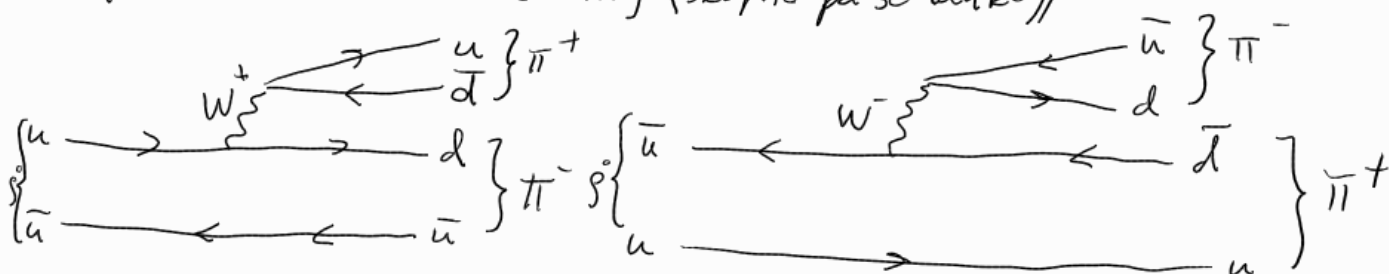


c) Prek šibke neutralne interakcije



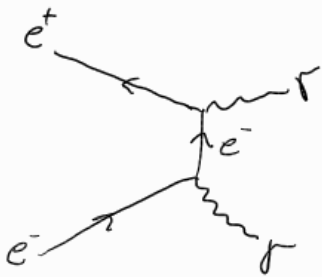
d) Prek šibke nabite interakcije

(okus v verteksih se ne dranja (skupno pa se lahko))

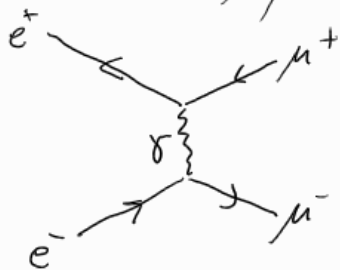


2) $e^+e^- \rightarrow \gamma\gamma$ (anihilacija)

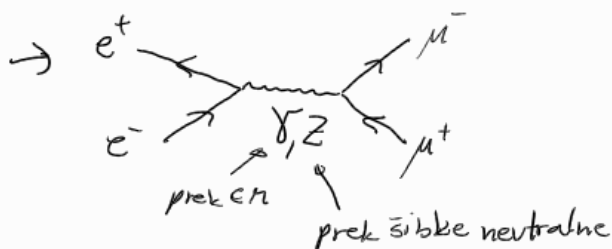
• EM interakcija



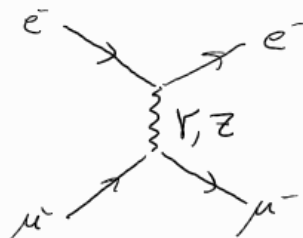
3) $e^+e^- \rightarrow \mu^-\mu^+$



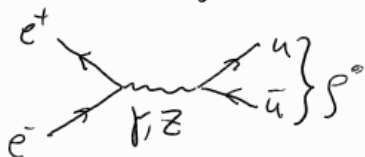
Ni ok, ker se v verteksih (vozliščih) ne ohranja okus



4) sipanje e^- in μ^- : $e^-\mu^- \rightarrow e^-\mu^-$

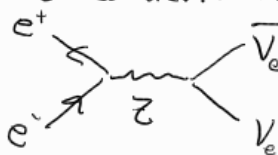


5) $e^+e^- \rightarrow \rho^0(u\bar{u})$



6) $e^+e^- \rightarrow \nu_e \bar{\nu}_e$

• šibka neutralna



7) $p e^- \rightarrow n \bar{\nu}_e$

$\downarrow$ $\downarrow$
uud udd

• proces ni mogoč, ker krši ohranitev leptonskega št. (+1 na levi, -1 na desni)

8) $K^+ \rightarrow \mu^+ \nu_\mu$; $m_{K^+}c^2 = 494 \text{ MeV}$, $m_{\mu^+}c^2 = 106 \text{ MeV}$, $m_{\nu_\mu}c^2 = 0$

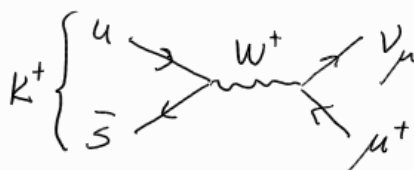
$\downarrow$ $\downarrow$
u $\bar{s}$ $m_{K^+}c^2 > m_{\mu^+}c^2 + m_{\nu_\mu}c^2 \checkmark$ (energija) • skus se ne ohranja $\Rightarrow$ šibka nabita int.

$e_c = e_0 \checkmark$ (naboj)

$B_L = 0 = B_D \checkmark$ (barionsko št.)

$L_L = 0 = L_D \checkmark$ (leptonske št.)

$L_{\mu L} = 0 = L_{\mu D} \checkmark$ (mionska Lept. št.)



$$9) \pi^0 \rightarrow K^+ e^- \bar{\nu}_e$$

• proces ni mogoč, ker $m_{K^+} c^2 > m_{\pi^0} c^2$

$$10) \Lambda^0 \rightarrow K^+ K^-$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ uds & u\bar{s} & s\bar{u} \end{array}$$

• proces ni mogoč, ker krši ohranitev barionskega št. ($B_L=1, B_D=0$)

$$11) B^0 \rightarrow D^- D^+ \quad m_{B^0} c^2 = 5,28 \text{ GeV}, m_{D^\pm} c^2 = 1,87 \text{ GeV}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ d\bar{b} & d\bar{c} & c\bar{d} \end{array}$$

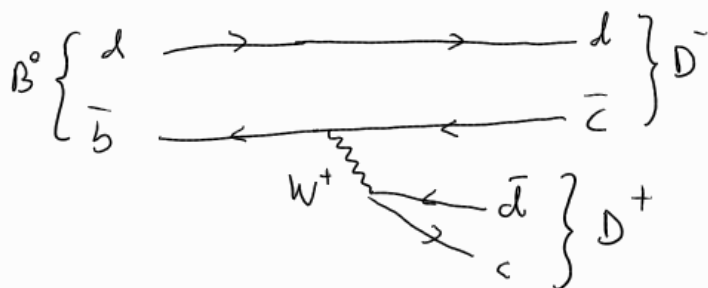
• okus se NE ohranja $\Rightarrow$ nabita šibka int.

$$m_{B^0} c^2 > 2m_{D^\pm} c^2 \quad \checkmark$$

$$e_L = e_D \quad \checkmark$$

$$B_L = 0 = B_D \quad \checkmark$$

$$\text{ni leptonov} \Rightarrow L_L = L_D = 0 \quad \checkmark$$



$\Rightarrow$ proces je mogoč